

Testing Plan

1. Mechanical Stability & Surface Interaction

Sliding Test

- Does the keyboard slide during normal use?
- How much force is required to initiate movement?
- Does temperature affect the grip performance?
- What material performs best for the feet (rubber, silicone, carpet-texture, etc.)?

Impact & Structural Testing

- Does anything break if excessive force is applied to a key?
 - Does the casing withstand accidental impacts (e.g., being hit by a cup)?
 - When shaken, does anything come loose internally?
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2. Electrical Validation

Voltage Testing

- Is sufficient voltage supplied to the PCB?
 - Does the voltage drop when a single key is pressed?
 - Does the voltage drop when multiple keys are pressed simultaneously?
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3. Functional Reliability

Key Registration

- Does each key register correctly when pressed?
- When multiple keys are pressed simultaneously, do all inputs register correctly?
- Is there any ghosting or missed input under stress conditions?

Cable & Port Reliability

- Does repeated plugging and unplugging of the USB cable cause failure?
 - Does the connector show signs of mechanical stress over time?
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4. Thermal Considerations

- Does heat buildup occur during extended use?
 - Is heat dissipation sufficient to prevent long-term component damage?
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5. Future Iterations

- What design changes would improve mechanical durability?
- How could electrical reliability be improved?
- What testing should be performed earlier in the development process?
- How could automate this with a python or arduino script?